

Set Notation for the Cross-Method Saliency Framework

Purpose

This note defines the set-theoretic notation behind the cross-method saliency framework used for SMS spam attribution analysis. The notation itself is standard; the novelty lies in what the sets represent and how their overlap and disagreement are interpreted.

Universe and attribution sets

Let

$$U = \{\text{all real tokens in a given message}\}$$

be the universe of discourse for one message.

Let

$$A \subseteq U$$

be the set of tokens selected as salient by the attention-based attribution method, and let

$$R \subseteq U$$

be the set of tokens selected as salient by the robust or classical attribution pool.

Core decomposition

The message-level saliency structure is decomposed into three regions.

1. Shared set

$$S = A \cap R$$

This is the **overlap** or **agreement** set: tokens selected by both attention and the robust/classical methods.

2. Attention-exclusive set

$$E_A = A \setminus R = \{x \mid x \in A \text{ and } x \notin R\}$$

This is the part of A that remains after removing tokens also found in R . In plain language, it is the set of tokens selected by attention **but not** by the robust/classical pool.

3. Robust-exclusive set

$$E_R = R \setminus A = \{x \mid x \in R \text{ and } x \notin A\}$$

This is the part of R that remains after removing tokens also found in A . In plain language, it is the set of tokens selected by the robust/classical pool **but not** by attention.

How “without” works formally

In this framework, the word *without* means **set difference**. Thus,

$$A \setminus R$$

is read as “A minus R,” “A without R,” or “A setminus R.”

Set difference is related to complement by

$$A \setminus R = A \cap R^c,$$

where R^c denotes the complement of R relative to the universe U . This is why “without” can feel like a kind of constrained negation: it is the part of one set that survives after intersecting with the complement of the other.

Agreement and disagreement

The framework studies both convergence and divergence.

- **Agreement / convergence:** $A \cap R$
- **Disagreement / divergence:** the exclusive regions $A \setminus R$ and $R \setminus A$

A useful aggregate disagreement object is the **symmetric difference**:

$$A \Delta R = (A \setminus R) \cup (R \setminus A).$$

This is the set of tokens that appear in exactly one of the two attribution sets.

Interpretive role in the project

Under this notation, the framework can be read as follows:

- $A \cap R$ identifies tokens on which mechanistically different methods converge.
- $A \setminus R$ isolates attention-exclusive salience, that is, tokens attention prioritizes that are not simultaneously prioritized by the robust/classical pool.

- $R \setminus A$ isolates robust-exclusive salience, that is, tokens emphasized by classical or robust methods but not by attention.

This makes the framework suitable for separating **shared lexical salience** from **residual disagreement**, where the latter may be analytically interesting even when the former is mostly explained by obvious surface cues.

Spoken readout for presentation

A clear way to say the notation aloud is:

- $A \cap R$: “A intersect R” or “the overlap set”
- $A \setminus R$: “A minus R” or “attention-only”
- $R \setminus A$: “R minus A” or “robust-only”
- $A \triangle R$: “A symmetric-difference R” or “the total disagreement set”

For oral presentation, the cleanest phrasing is often: *shared tokens, attention-only tokens, and robust-only tokens*.

Compact paper-ready definition

A concise version suitable for a thesis or paper is:

Let U denote the set of tokens in a message. Let $A \subseteq U$ be the set of tokens selected by the attention-based attribution method, and let $R \subseteq U$ be the set of tokens selected by the robust/classical attribution pool. We decompose message-level salience into the shared set $A \cap R$, the attention-exclusive set $A \setminus R$, and the robust-exclusive set $R \setminus A$. Their union covers the attributed region $A \cup R$, while the symmetric difference $(A \setminus R) \cup (R \setminus A)$ captures total disagreement between the two attribution families.